Government of Canada Gouvernement du Canada

V-17—Temperature correction of provers

Category: Volume
Issue date: 2019-12-19
Effective date: 2020-04-01
Revision number: 3
Supersedes: V-17 (rev. 2)

Table of contents

- 1.0 Scope
- 2.0 Purpose
- 3.0 References
- 4.0 Background
- 5.0 Policy
- 6.0 Calculations
- 7.0 Revisions
- Appendix: Correction of the indicated volume of provers at test temperature

1.0 Scope

This bulletin applies to open narrow neck local standards (provers) which may be used to calibrate or certify volumetric measuring devices or other provers.

Exemption phase-out

Previous versions of this bulletin exempted provers with a nominal capacity equal to or less than 50 L. That is, prover shell volume changes due to temperature were simply ignored for these smaller provers as the volume change due to temperature is usually relatively small. Recently, Measurement Canada has decided that these calculations must be made for smaller provers as well. While it is true that the changes are small when the temperature is moderate, when proving at very high or very low temperatures, the shell

correction can account for a significant portion of the applicable limit of error for the meter being tested. As of April 1, 2020, prover shell corrections must be made for all open neck provers regardless of the nominal capacity. These calculations must therefore be made for 20 L provers or test measures. It is suggested that inspectors and authorized service providers immediately begin the transition to the complete test.

2.0 Purpose

The purpose of this bulletin is to define the policy relating to the correction of a prover's indicated volume to account for volume changes due to expansion or contraction of the material used in construction when the actual temperature is different than the reference temperature.

3.0 References

The tables in the Appendix provide the volume correction factors (VCFs) for various standard size provers over a wide range of temperatures. For more information, refer to the following documents:

- Measurement Canada's Guidelines for the Calibration and Certification of Volumetric Standards;
- Field Inspection Manual—Volumetric Measuring Devices, Part 3, section 23.

4.0 Background

Provers are calibrated to a specific volume at a reference temperature, which is usually 15 °C for narrow neck provers used for petroleum and chemical service. If a prover is used at any temperature other than the reference temperature, a correction is required to address the expansion or contraction of the material used to construct the prover.

It is important to understand that a prover’s volume changes with temperature and this change is different for various materials. For this reason, the prover's true volume must be calculated based on the operating temperature. It also means that when comparing provers constructed of different materials, a correction must be applied to both provers, even if the temperature of each is the same. For example, stainless steel has a cubical coefficient of thermal expansion of 0.0000518/°C, which is about one and a half times that of mild steel at 0.0000335/°C. Therefore, the volume of the stainless steel prover will

change more significantly than that of a mild steel prover as the operating temperature moves away from the reference temperature. This correction is referred to as the correction for the temperature of the steel (Cts) of the meter (m) or prover (p). *In* this acronym, steel is a general term used to represent the material of construction of the meter or prover.

Measurement Canada considers it essential to correct the volume of provers according to temperature when testing meters. While the correction may be made using a formula and the appropriate coefficient for the prover, in some cases it may be simpler to refer to the specific correction table in the Appendix.

5.0 Policy

Measurement Canada inspectors and service providers authorized by Measurement Canada to inspect devices must correct the volume of all provers to the appropriate reference temperature (e.g., 15 °C) when conducting meter inspections or examinations using in-service or acceptance limits of error.

> **Note:** This is a change from the previous policy that exempted provers of less than 50 L.

Corrections must always be based upon the actual cubical coefficient of thermal expansion applicable to the material used to fabricate the prover shell. If this information is not available on the prover data plate or accompanying documentation, please contact Measurement Canada for guidance in determining an appropriate coefficient.

Corrections must be made to the nearest 0.1 °C. As the included tables show corrections to the nearest 1.0 °C, linear interpolation must be used to determine the correction when the temperature is not a whole number. This additional step is not required if using the formula.

> **Note:** Interpolation of the temperature is not required when the correction would be less than the resolution of the prover's gauge plate. This may be the case especially with smaller provers and small temperature changes. Rounding may be applied for calculated correction values less than the resolution of the prover's gauge plate.

6.0 Calculations

To calculate the change in prover volume due to expansion or contraction of the prover shell, you must know the cubical coefficient of thermal expansion per degree Celsius for the material used to construct the prover. This data is available from the manufacturer and is usually stamped on the prover's data plate. If the information is not readily available, please contact Measurement Canada.

> **Note:** In some cases, the cubical coefficient of thermal expansion may be identified with Greek letters, typically alpha (α) or gamma (γ). Table 1 contains some commonly accepted values.

Table 1: Cubical coefficient of thermal expansion for some common materials

Material	Cubical coefficient of thermal expansion / °C
Mild steel	0.000 033 5 or 3.35×10^{-5}
Stainless steel type 304 (18/8)	0.000 051 8 or 5.18×10^{-5}
Stainless steel type 316	0.000 045 4 or 4.54×10^{-5}
Stainless steel type 17.4	0.000 032 4 or 3.24×10^{-5}
Borosilicate glassware (Pyrex®, Bomex®, etc.)	0.000 010 0 or 1.00×10^{-5}

> **Note:** The base or reference temperature that was used to calibrate the prover (usually 15 °C) and the prover shell temperature at the time of using (reading) the prover must be known before performing the calculations.

> **Note:** Borosilicate glassware such as Pyrex® and Bomex® is usually calibrated at a reference temperature of 20 °C.

> **Note:** Sanitary provers for use with milk are usually calibrated at a reference temperature of 4.4 °C and are exclusively built of stainless steel.

The prover shell temperature is usually determined using a thermometer mounted in a thermowell in the prover body. Although the temperature of the liquid in the vessel is being sensed, once stabilized, it is deemed to be representative of the prover shell temperature. The formula is the following:

$$Ctsp = (Tp - Tref) \times CCE \times Vref$$

Where:

- Ctsp = correction for the thermal expansion/contraction of the prover in the same units as the prover volume
- Tp = temperature, in Celsius, of the prover when the gauge plate is read
- Tref = reference temperature at which the prover was calibrated (usually 15 °C)
- CCE = cubical coefficient of thermal expansion per degree Celsius for the material used to construct the prover (shell material)
- Vref = certified base volume of the prover at the reference temperature

> **Note:** As 20 L field test measures are not normally fitted with thermowells or thermometers and it is generally not advisable to place electronic thermometer probes directly into the product, shell corrections for 20 L test measures may be made using the same average temperature as used for the automatic temperature compensation calculation. This is only acceptable if the gauge plate is read and the correction is performed immediately after filling the measure.

Alternate calculation

The tables in the Appendix include a volume correction factor column. This factor is the multiplier to be used to calculate the true volume of the prover at a given temperature. The true corrected volume of a prover, at the zero mark on the gauge plate, is equal to the indicated volume multiplied by the factor at the observed temperature.

In the examples below, the full calculation is shown first, followed by the alternate calculation. This is done in order to show how the results compare between the two methods. The result is the corrected volume in the prover (CVp).

Examples

1. The gauge plate reading on a 20 L type 304 stainless steel prover shell read at 0 °C must be adjusted as follows:

$$CVp = (0\ °C - 15\ °C) \times 0.0000518\ /\ °C \times 20\ L = -0.016\ L\ (-16\ mL)$$

In order to get the correct value, you must subtract 0.016 L from the prover reading. Therefore, if the gauge plate reads exactly 20 L at 0 °C, then the true volume in the prover is CVp = 20 L − 0.016 L = 19.984 L.

Alternate calculation using a VCF

$$CVp = 20\ L \times 0.99922 = 19.9844\ L$$

> **Note:** a 20 litre prover typically has 10 mL graduations. At best, these can be read to 1/2 graduation so corrections of less than this value must be rounded off appropriately. In the example above, 19.984 L is effectively rounded to 19.98 L or perhaps to 19.985 L. This means small temperature differences from 15 °C have a negligible effect on the prover shell and may be ignored.

2. The gauge plate reading on a 1000 L mild steel prover shell read at 20 °C must be adjusted as follows:

$$CVp = (20\ °C - 15\ °C) \times 0.0000335\ °C \times 1000\ L = +0.168\ L\ (168\ mL)$$

In order to get the correct value, you must add 0.168 L (168 mL) to the prover reading. Therefore, if the gauge plate reads 999.832 L at 20 °C, then the true volume in the prover is 999.832 L + 0.168 L = 1000 L.

Alternate calculation using VCF

$$CVp = 999.832\ L \times 1.00017 = 1000.00\ L$$

3. The gauge plate reading on a 2500 L type 304 stainless steel prover shell read at −12 °C must be adjusted as follows:

$$CVp = (-12\ °C - 15\ °C) \times 0.0000518\ °C \times 2500L = -3.497\ L$$

In order to get the correct value, you must subtract 3.497 L from the prover reading. Therefore, if the gauge plate reads exactly 2500 L at −12 °C, then the true volume in the prover is 2500 L − 3.497 L = 2496.50 L.

Alternate calculation using a VCF

$$CVp = 2500\ L \times 0.99860\ = 2496.5\ L$$

Clarification

- Above 15 °C, the actual volume in the prover is greater than the volume indicated on the gauge plate, so you must add to the indicated volume.
- Below 15 °C, the actual volume in the prover is less than the volume indicated on the gauge plate, so you must subtract from the indicated volume.

7.0 Revisions

The purpose of revision 3 was to:

- make the cubical coefficient of thermal expansion calculation applicable to all prover shells, including those of less than 50 L;
- add an example of a calculation for a 20 L test measure.

The purpose of revision 2 was to:

- add common prover sizes;
- expand tables to cover temperatures from −30 °C to 40 °C;
- make separate tables for mild steel and stainless steel type 304 (18/8) provers;
- add a column with VCFs;
- make editorial and format changes;
- add a section dealing with selecting and calculating appropriate corrections.

The purpose of revision 1 was to:

- revise the bulletin format;
- make minor editorial changes;
- remove the requirement for inspectors to annotate the inspection certificate to show that the prover was corrected for temperature;
- define the physical size and use condition when volumetric standards require volume correction for the temperature of the shell and include a column for 1500 L stainless steel prover corrections.

Appendix: Correction of the indicated volume of provers at test temperature

Note: The corrections in the tables below are only valid for provers with a reference temperature of 15 °C and are indicated in litres rounded off to the nearest 0.001 L (1 mL). In all cases, the indicated correction, at observed temperature, is added to the prover indication.

Note: Interpolation for temperature is permitted. Rounding for resolution is also permitted.

Table A.1: Correction of the indicated volume of type 304 or 18/8 stainless steel provers at test temperature
Cubical coefficient of thermal expansion = 0.0000518 /°C

Prover shell temperature in degrees Celsius	Ctsp	Volume in litres to be added to correct observed prover reading for various nominal prover sizes (20 L to 2500 L)							
		20 L	50 L	100 L	250 L	500 L	1000 L	1500 L	2500 L
−30.0	0.99767	−0.047	−0.117	−0.233	−0.583	−1.166	−2.331	−3.497	−5.828
−29.0	0.99772	−0.046	−0.114	−0.228	−0.570	−1.140	−2.279	−3.419	−5.698
−28.0	0.99777	−0.045	−0.111	−0.223	−0.557	−1.114	−2.227	−3.341	−5.569
−27.0	0.99782	−0.044	−0.109	−0.218	−0.544	−1.088	−2.176	−3.263	−5.439
−26.0	0.99788	−0.042	−0.106	−0.212	−0.531	−1.062	−2.124	−3.186	−5.310
−25.0	0.99793	−0.041	−0.104	−0.207	−0.518	−1.036	−2.072	−3.108	−5.180
−24.0	0.99798	−0.040	−0.101	−0.202	−0.505	−1.010	−2.020	−3.030	−5.051
−23.0	0.99803	−0.039	−0.098	−0.197	−0.492	−0.984	−1.968	−2.953	−4.921
−22.0	0.99808	−0.038	−0.096	−0.192	−0.479	−0.958	−1.917	−2.875	−4.792
−21.0	0.99814	−0.037	−0.093	−0.186	−0.466	−0.932	−1.865	−2.797	−4.662

Prover shell temperature in degrees Celsius	Ctsp	Volume in litres to be added to correct observed prover reading for various nominal prover sizes (20 L to 2500 L)							
		20 L	50 L	100 L	250 L	500 L	1000 L	1500 L	2500 L
−20.0	0.99819	−0.036	−0.091	−0.181	−0.453	−0.907	−1.813	−2.720	−4.533
−19.0	0.99824	−0.035	−0.088	−0.176	−0.440	−0.881	−1.761	−2.642	−4.403
−18.0	0.99829	−0.034	−0.085	−0.171	−0.427	−0.855	−1.709	−2.564	−4.274
−17.0	0.99834	−0.033	−0.083	−0.166	−0.414	−0.829	−1.658	−2.486	−4.144
−16.0	0.99839	−0.032	−0.080	−0.161	−0.401	−0.803	−1.606	−2.409	−4.015
−15.0	0.99845	−0.031	−0.078	−0.155	−0.389	−0.777	−1.554	−2.331	−3.885
−14.0	0.99850	−0.030	−0.075	−0.150	−0.376	−0.751	−1.502	−2.253	−3.756
−13.0	0.99855	−0.029	−0.073	−0.145	−0.363	−0.725	−1.450	−2.176	−3.626
−12.0	0.99860	−0.028	−0.070	−0.140	−0.350	−0.699	−1.399	−2.098	−3.497
−11.0	0.99865	−0.027	−0.067	−0.135	−0.337	−0.673	−1.347	−2.020	−3.367
−10.0	0.99871	−0.026	−0.065	−0.130	−0.324	−0.648	−1.295	−1.943	−3.238
−9.0	0.99876	−0.025	−0.062	−0.124	−0.311	−0.622	−1.243	−1.865	−3.108
−8.0	0.99881	−0.024	−0.060	−0.119	−0.298	−0.596	−1.191	−1.787	−2.979
−7.0	0.99886	−0.023	−0.057	−0.114	−0.285	−0.570	−1.140	−1.709	−2.849
−6.0	0.99891	−0.022	−0.054	−0.109	−0.272	−0.544	−1.088	−1.632	−2.720
−5.0	0.99896	−0.021	−0.052	−0.104	−0.259	−0.518	−1.036	−1.554	−2.590
−4.0	0.99902	−0.020	−0.049	−0.098	−0.246	−0.492	−0.984	−1.476	−2.461
−3.0	0.99907	−0.019	−0.047	−0.093	−0.233	−0.466	−0.932	−1.399	−2.331
−2.0	0.99912	−0.018	−0.044	−0.088	−0.220	−0.440	−0.881	−1.321	−2.202
−1.0	0.99917	−0.017	−0.041	−0.083	−0.207	−0.414	−0.829	−1.243	−2.072
0.0	0.99922	−0.016	−0.039	−0.078	−0.194	−0.389	−0.777	−1.166	−1.943

Prover shell temperature in degrees Celsius	Ctsp	Volume in litres to be added to correct observed prover reading for various nominal prover sizes (20 L to 2500 L)							
		20 L	50 L	100 L	250 L	500 L	1000 L	1500 L	2500 L
1.0	0.99927	−0.015	−0.036	−0.073	−0.181	−0.363	−0.725	−1.088	−1.813
2.0	0.99933	−0.013	−0.034	−0.067	−0.168	−0.337	−0.673	−1.010	−1.684
3.0	0.99938	−0.012	−0.031	−0.062	−0.155	−0.311	−0.622	−0.932	−1.554
4.0	0.99943	−0.011	−0.028	−0.057	−0.142	−0.285	−0.570	−0.855	−1.425
5.0	0.99948	−0.010	−0.026	−0.052	−0.130	−0.259	−0.518	−0.777	−1.295
6.0	0.99953	−0.009	−0.023	−0.047	−0.117	−0.233	−0.466	−0.699	−1.166
7.0	0.99959	−0.008	−0.021	−0.041	−0.104	−0.207	−0.414	−0.622	−1.036
8.0	0.99964	−0.007	−0.018	−0.036	−0.091	−0.181	−0.363	−0.544	−0.907
9.0	0.99969	−0.006	−0.016	−0.031	−0.078	−0.155	−0.311	−0.466	−0.777
10.0	0.99974	−0.005	−0.013	−0.026	−0.065	−0.130	−0.259	−0.389	−0.648
11.0	0.99979	−0.004	−0.010	−0.021	−0.052	−0.104	−0.207	−0.311	−0.518
12.0	0.99984	−0.003	−0.008	−0.016	−0.039	−0.078	−0.155	−0.233	−0.389
13.0	0.99990	−0.002	−0.005	−0.010	−0.026	−0.052	−0.104	−0.155	−0.259
14.0	0.99995	−0.001	−0.003	−0.005	−0.013	−0.026	−0.052	−0.078	−0.130
15.0	1.00000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
16.0	1.00005	0.001	0.003	0.005	0.013	0.026	0.052	0.078	0.130
17.0	1.00010	0.002	0.005	0.010	0.026	0.052	0.104	0.155	0.259
18.0	1.00016	0.003	0.008	0.016	0.039	0.078	0.155	0.233	0.389
19.0	1.00021	0.004	0.010	0.021	0.052	0.104	0.207	0.311	0.518
20.0	1.00026	0.005	0.013	0.026	0.065	0.130	0.259	0.389	0.648
21.0	1.00031	0.006	0.016	0.031	0.078	0.155	0.311	0.466	0.777

Prover shell temperature in degrees Celsius	Ctsp	Volume in litres to be added to correct observed prover reading for various nominal prover sizes (20 L to 2500 L)							
		20 L	50 L	100 L	250 L	500 L	1000 L	1500 L	2500 L
22.0	1.00036	0.007	0.018	0.036	0.091	0.181	0.363	0.544	0.907
23.0	1.00041	0.008	0.021	0.041	0.104	0.207	0.414	0.622	1.036
24.0	1.00047	0.009	0.023	0.047	0.117	0.233	0.466	0.699	1.166
25.0	1.00052	0.010	0.026	0.052	0.130	0.259	0.518	0.777	1.295
26.0	1.00057	0.011	0.028	0.057	0.142	0.285	0.570	0.855	1.425
27.0	1.00062	0.012	0.031	0.062	0.155	0.311	0.622	0.932	1.554
28.0	1.00067	0.013	0.034	0.067	0.168	0.337	0.673	1.010	1.684
29.0	1.00073	0.015	0.036	0.073	0.181	0.363	0.725	1.088	1.813
30.0	1.00078	0.016	0.039	0.078	0.194	0.389	0.777	1.166	1.943
31.0	1.00083	0.017	0.041	0.083	0.207	0.414	0.829	1.243	2.072
32.0	1.00088	0.018	0.044	0.088	0.220	0.440	0.881	1.321	2.202
33.0	1.00093	0.019	0.047	0.093	0.233	0.466	0.932	1.399	2.331
34.0	1.00098	0.020	0.049	0.098	0.246	0.492	0.984	1.476	2.461
35.0	1.00104	0.021	0.052	0.104	0.259	0.518	1.036	1.554	2.590
36.0	1.00109	0.022	0.054	0.109	0.272	0.544	1.088	1.632	2.720
37.0	1.00114	0.023	0.057	0.114	0.285	0.570	1.140	1.709	2.849
38.0	1.00119	0.024	0.060	0.119	0.298	0.596	1.191	1.787	2.979
39.0	1.00124	0.025	0.062	0.124	0.311	0.622	1.243	1.865	3.108
40.0	1.00130	0.026	0.065	0.130	0.324	0.648	1.295	1.943	3.238

Table A.2: Correction of the indicated volume of mild steel provers (all types)
Cubical coefficient of thermal expansion = 0.0000335 /°C

Prover shell temperature in degrees Celcius	Ctsp	Volume in litres to be added to correct observed prover indication for various nominal prover sizes (20 L to 2500 L)							
		20 L	50 L	100 L	250 L	500 L	1000 L	1500 L	2500 L
−30.0	0.99849	−0.030	−0.075	−0.151	−0.377	−0.754	−1.508	−2.261	−3.769
−29.0	0.99853	−0.029	−0.074	−0.147	−0.369	−0.737	−1.474	−2.211	−3.685
−28.0	0.99856	−0.029	−0.072	−0.144	−0.360	−0.720	−1.441	−2.161	−3.601
−27.0	0.99859	−0.028	−0.070	−0.141	−0.352	−0.704	−1.407	−2.111	−3.518
−26.0	0.99863	−0.027	−0.069	−0.137	−0.343	−0.687	−1.374	−2.060	−3.434
−25.0	0.99866	−0.027	−0.067	−0.134	−0.335	−0.670	−1.340	−2.010	−3.350
−24.0	0.99869	−0.026	−0.065	−0.131	−0.327	−0.653	−1.307	−1.960	−3.266
−23.0	0.99873	−0.025	−0.064	−0.127	−0.318	−0.637	−1.273	−1.910	−3.183
−22.0	0.99876	−0.025	−0.062	−0.124	−0.310	−0.620	−1.240	−1.859	−3.099
−21.0	0.99879	−0.024	−0.060	−0.121	−0.302	−0.603	−1.206	−1.809	−3.015
−20.0	0.99883	−0.023	−0.059	−0.117	−0.293	−0.586	−1.173	−1.759	−2.931
−19.0	0.99886	−0.023	−0.057	−0.114	−0.285	−0.570	−1.139	−1.709	−2.848
−18.0	0.99889	−0.022	−0.055	−0.111	−0.276	−0.553	−1.106	−1.658	−2.764
−17.0	0.99893	−0.021	−0.054	−0.107	−0.268	−0.536	−1.072	−1.608	−2.680
−16.0	0.99896	−0.021	−0.052	−0.104	−0.260	−0.519	−1.039	−1.558	−2.596
−15.0	0.99900	−0.020	−0.050	−0.101	−0.251	−0.503	−1.005	−1.508	−2.513
−14.0	0.99903	−0.019	−0.049	−0.097	−0.243	−0.486	−0.972	−1.457	−2.429
−13.0	0.99906	−0.019	−0.047	−0.094	−0.235	−0.469	−0.938	−1.407	−2.345
−12.0	0.99910	−0.018	−0.045	−0.090	−0.226	−0.452	−0.905	−1.357	−2.261
−11.0	0.99913	−0.017	−0.044	−0.087	−0.218	−0.436	−0.871	−1.307	−2.178
−10.0	0.99916	−0.017	−0.042	−0.084	−0.209	−0.419	−0.838	−1.256	−2.094

Prover shell temperature in degrees Celcius	Ctsp	Volume in litres to be added to correct observed prover indication for various nominal prover sizes (20 L to 2500 L)							
		20 L	50 L	100 L	250 L	500 L	1000 L	1500 L	2500 L
−9.0	0.99920	−0.016	−0.040	−0.080	−0.201	−0.402	−0.804	−1.206	−2.010
−8.0	099923	−0.015	−0.039	−0.077	−0.193	−0.385	−0.771	−1.156	−1.926
−7.0	0.99926	−0.015	−0.037	−0.074	−0.184	−0.369	−0.737	−1.106	−1.843
−6.0	0.99930	−0.014	−0.035	−0.070	−0.176	−0.352	−0.704	−1.055	−1.759
−5.0	0.99933	−0.013	−0.034	−0.067	−0.168	−0.335	−0.670	−1.005	−1.675
−4.0	0.99936	−0.013	−0.032	−0.064	−0.159	−0.318	−0.637	−0.955	−1.591
−3.0	0.99940	−0.012	−0.030	−0.060	−0.151	−0.302	−0.603	−0.905	−1.508
−2.0	0.99943	−0.011	−0.028	−0.057	−0.142	−0.285	−0.570	−0.854	−1.424
−1.0	0.99946	−0.011	−0.027	−0.054	−0.134	−0.268	−0.536	−0.804	−1.340
0.0	0.99950	−0.010	−0.025	−0.050	−0.126	−0.251	−0.503	−0.754	−1.256
1.0	0.99953	−0.009	−0.023	−0.047	−0.117	−0.235	−0.469	−0.704	−1.173
2.0	0.99956	−0.009	−0.022	−0.044	−0.109	−0.218	−0.436	−0.653	−1.089
3.0	0.99960	−0.008	−0.020	−0.040	−0.101	−0.201	−0.402	−0.603	−1.005
4.0	0.99963	−0.007	−0.018	−0.037	−0.092	−0.184	−0.369	−0.553	−0.921
5.0	0.99967	−0.007	−0.017	−0.034	−0.084	−0.168	−0.335	−0.503	−0.838
6.0	0.99970	−0.006	−0.015	−0.030	−0.075	−0.151	−0.302	−0.452	−0.754
7.0	0.99973	−0.005	−0.013	−0.027	−0.067	−0.134	−0.268	−0.402	−0.670
8.0	0.99977	−0.005	−0.012	−0.023	−0.059	−0.117	−0.235	−0.352	−0.586
9.0	0.99980	−0.004	−0.010	−0.020	−0.050	−0.101	−0.201	−0.302	−0.503
10.0	0.99983	−0.003	−0.008	−0.017	−0.042	−0.084	−0.168	−0.251	−0.419
11.0	0.99987	−0.003	−0.007	−0.013	−0.034	−0.067	−0.134	−0.201	−0.335

Prover shell temperature in degrees Celcius	Ctsp	Volume in litres to be added to correct observed prover indication for various nominal prover sizes (20 L to 2500 L)							
		20 L	50 L	100 L	250 L	500 L	1000 L	1500 L	2500 L
12.0	0.99990	−0.002	−0.005	−0.010	−0.025	−0.050	−0.101	−0.151	−0.251
13.0	0.99993	−0.001	−0.003	−0.007	−0.017	−0.034	−0.067	−0.101	−0.168
14.0	0.99997	−0.001	−0.002	−0.003	−0.008	−0.017	−0.034	−0.050	−0.084
15.0	1.00000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
16.0	1.00003	0.001	0.002	0.003	0.008	0.017	0.034	0.050	0.084
17.0	1.00007	0.001	0.003	0.007	0.017	0.034	0.067	0.101	0.168
18.0	1.00010	0.002	0.005	0.010	0.025	0.050	0.101	0.151	0.251
19.0	1.00013	0.003	0.007	0.013	0.034	0.067	0.134	0.201	0.335
20.0	1.00017	0.003	0.008	0.017	0.042	0.084	0.168	0.251	0.419
21.0	1.00020	0.004	0.010	0.020	0.050	0.101	0.201	0.302	0.503
22.0	1.00023	0.005	0.012	0.023	0.059	0.117	0.235	0.352	0.586
23.0	1.00027	0.005	0.013	0.027	0.067	0.134	0.268	0.402	0.670
24.0	1.00030	0.006	0.015	0.030	0.075	0.151	0.302	0.452	0.754
25.0	1.00034	0.007	0.017	0.034	0.084	0.168	0.335	0.503	0.838
26.0	1.00037	0.007	0.018	0.037	0.092	0.184	0.369	0.553	0.921
27.0	1.00040	0.008	0.020	0.040	0.101	0.201	0.402	0.603	1.005
28.0	1.00044	0.009	0.022	0.044	0.109	0.218	0.436	0.653	1.089
29.0	1.00047	0.009	0.023	0.047	0.117	0.235	0.469	0.704	1.173
30.0	1.00050	0.010	0.025	0.050	0.126	0.251	0.503	0.754	1.256
31.0	1.00054	0.011	0.027	0.054	0.134	0.268	0.536	0.804	1.340
32.0	1.00057	0.011	0.028	0.057	0.142	0.285	0.570	0.854	1.424

Prover shell temperature in degrees Celcius	Ctsp	Volume in litres to be added to correct observed prover indication for various nominal prover sizes (20 L to 2500 L)							
		20 L	50 L	100 L	250 L	500 L	1000 L	1500 L	2500 L
33.0	1.00060	0.012	0.030	0.060	0.151	0.302	0.603	0.905	1.508
34.0	1.00064	0.013	0.032	0.064	0.159	0.318	0.637	0.955	1.591
35.0	1.00067	0.013	0.034	0.067	0.168	0.335	0.670	1.005	1.675
36.0	1.00070	0.014	0.035	0.070	0.176	0.352	0.704	1.055	1.759
37.0	1.00074	0.015	0.037	0.074	0.184	0.369	0.737	1.106	1.843
38.0	1.00077	0.015	0.039	0.077	0.193	0.385	0.771	1.156	1.926
39.0	1.00080	0.016	0.040	0.080	0.201	0.402	0.804	1.206	2.010
40.0	1.00084	0.017	0.042	0.084	0.209	0.419	0.838	1.256	2.094

Date modified:
2020-02-11